Lab 3 Calculations

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3.A RMS Voltage calculation

3.A.1 Sine wave

$$\begin{aligned} V_{\text{magnitude}} = V_{\text{peak-to-peak}} \times \frac{1}{2} &= V_{\text{RMS}} \times \sqrt{2} \\ 10V \times \frac{1}{2} &= V_{\text{RMS}} \times \sqrt{2} \\ V_{\text{RMS}} = \frac{5}{\sqrt{2}}V \approx 3.54V \end{aligned}$$

3.A.2 Square wave

$$\begin{aligned} V_{\text{magnitude}} = V_{\text{peak-to-peak}} \times \frac{1}{2} &= V_{\text{RMS}} \\ 10V \times \frac{1}{2} &= V_{\text{RMS}} \\ V_{\text{RMS}} = 5V \end{aligned}$$

3.A.3 Triangle wave

$$\begin{aligned} V_{\text{magnitude}} = V_{\text{peak-to-peak}} \times \frac{1}{2} &= V_{\text{RMS}} \times \sqrt{3} \\ 10V \times \frac{1}{2} &= V_{\text{RMS}} \times \sqrt{3} \\ V_{\text{RMS}} = \frac{5}{\sqrt{3}}V \approx 2.89V \end{aligned}$$

3.B Period calculation

3.B.1 100 Hz

$$\begin{aligned} \text{frequency} &= \frac{1}{\text{period}} \\ 100 \text{ Hz} &= \frac{1}{\text{period}} \\ \text{period} &= \frac{1}{100 \text{ Hz}} = 0.01s = 10ms \end{aligned}$$

3.B.2 1 kHz

$$\begin{aligned} \text{frequency} &= \frac{1}{\text{period}} \\ 1 \text{ kHz} = 1000 \text{ Hz} &= \frac{1}{\text{period}} \\ \text{period} &= \frac{1}{1000 \text{ Hz}} = 0.001s = 1ms \end{aligned}$$

3.B.3 100 kHz

$$\begin{aligned} \text{frequency} &= \frac{1}{\text{period}} \\ 100 \text{ kHz} = 100000 \text{ Hz} &= \frac{1}{\text{period}} \\ \text{period} &= \frac{1}{100000 \text{ Hz}} = 0.00001s = 10\mu s \end{aligned}$$